



FUNCTIONAL SPECIFICATION

Library Management System

A complete, plain-language description of how the system helps a library manage its collection, its members, and the lending of books — from the front desk or from anywhere.

Document

Functional Specification

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Audience

Prospective & current clients

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About this document

This specification describes **what** the system does and **how it behaves** for the people who use it. It deliberately avoids technical detail, so it can be read and reviewed by librarians, managers, and decision-makers without a technical background.

1 Introduction & Purpose

The Library Management System is a web-based application that helps a library run its entire day-to-day operation — its collection, its members, and the lending of books — from a single, clear, easy-to-use interface.

It replaces paper registers and scattered spreadsheets with one reliable, searchable system that is always up to date. Because it runs in an ordinary web browser, staff can use it from the front-desk computer, from other computers in the building, or — when appropriate — securely from anywhere.

This document presents the complete set of functions the system offers, written in plain language and organised by the everyday tasks a library performs.

2 Who Uses the System

The system is designed around the people who run the library each day.

- **Library staff (librarians).** The primary users. They manage the catalogue and members, record loans and returns, review what is overdue, and adjust the library's settings. All everyday work is done by staff on behalf of the library.
- **Members.** The people who borrow books. Each member is registered in the system so their borrowing can be tracked, but in this version members are served by staff at the desk rather than signing in themselves. (A member self-service option is noted under Future Enhancements.)

3 Key Concepts

A few simple terms are used throughout the system and this document.

Book (copy)	Every physical book on the shelf is recorded individually and given its own unique book number . Two copies of the same title are tracked as two separate books, so the library always knows exactly which copy is where.
Category	A subject or grouping — for example Fiction, History, or Children — that books are organised under to make browsing and searching easier.
Member	A person registered with the library who is allowed to borrow books.
Loan	A record of one book being borrowed by one member — from the moment it is taken out to the moment it is returned.
Loan status	Where a loan stands right now: Active , Overdue , or Returned .
Book status	Whether a copy is part of the collection: Active , or removed as Lost , Damaged , or Withdrawn .

4 Book Catalogue Management

Everything needed to keep the library's collection accurate and easy to browse.

- ◆ **Register a book** with its title, author, unique book number, category, publisher, and price.
- ◆ **Edit** any book's details at any time to keep records correct.
- ◆ **Browse the full catalogue** in a clear, sortable list.
- ◆ **Search and filter** the collection quickly — for example by book number, title, author, or category.
- ◆ **Remove a book from the collection**, recording **why** — Lost, Damaged, or Withdrawn — with an optional note. The book keeps its full history and is never erased.
- ◆ **Review removed books** on a dedicated archive screen showing the reason and note for each, filterable by reason.
- ◆ **Restore a book** that was removed in error or has since been found, returning it to the active collection.
- ◆ **Unique book numbers** are enforced across the active collection.

Removing a book

Removal always requires a reason, so the library can later distinguish “we lost twelve books” from “we retired twelve worn-out books”. An optional note records any further detail.

A book currently on loan cannot be removed until it is returned. Removed books do not appear in the catalogue, cannot be lent, and are not counted in the collection totals on the home screen — but they remain visible on the archive screen and in the borrowing history of any member who borrowed them.

Restoring a book

A removed book can be restored to the active collection at any time — for a mis-click, or when a lost book turns up. Restoring is always a deliberate action: editing a removed book's details never silently returns it to the collection.

Because a removed book's number becomes available for reuse, another book may have taken that number in the meantime. When this happens the system says so clearly and asks for a new number before completing the restore.

5 Category Management

Organise the collection into meaningful groups.

- ◆ **Create and edit categories**, each with a name and a short description.
- ◆ **Organise books** by assigning each one to a category.
- ◆ **Browse by category** to see how the collection is distributed.
- ◆ **Protected removal** — a category that still has books assigned to it cannot be removed by mistake; it is archived instead, keeping the collection consistent.
- ◆ **Unique category names** prevent confusing duplicates.

6 Member Management

Keep an accurate register of everyone entitled to borrow.

- ◆ **Register a member** with their name, email, address, and phone number.
- ◆ **Edit** member details as they change.
- ◆ **Browse, search, and filter** the member list — for example to find someone by name.
- ◆ **Archive a member** who is no longer active, while preserving their borrowing history.
- ◆ **Protected removal** — a member who currently has books on loan cannot be removed until those books are returned.
- ◆ **Unique member names** help avoid duplicate records.

7 Loans & Returns

The heart of the system: lending books out and taking them back, with sensible safeguards built in.

- ◆ **Lend a book** by selecting the member and the book being borrowed. The system creates a loan and records the date automatically.
- ◆ **Return a book** in a single step, which closes the loan and stamps the return date.
- ◆ **See all current loans** — who has what, and since when.
- ◆ **Filter loans by status** to focus on active, overdue, or completed loans.

Every loan moves through three clear stages:

Active Borrowed and still within the allowed lending period.	Overdue Kept beyond the allowed period — flagged automatically.	Returned Given back; the loan is closed and kept for history.
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Built-in safeguards

- ◆ A single book copy can be on **only one active loan at a time**, so the system never double-lends the same physical book.
- ◆ A member can hold **no more than the allowed number of books** at once (see System Settings).
- ◆ Only books in the **active collection** can be lent.

8 Overdue Management

The system keeps track of due dates so staff don't have to.

- ◆ **Automatic overdue detection** — each active loan is measured against the library's allowed lending period, and is marked **Overdue** on its own once that period passes.
- ◆ **Automatic correction** — if the library later extends its lending period, loans that are no longer late are returned to **Active** automatically.
- ◆ **Overdue at a glance** — staff can see every overdue loan in one place, making follow-up straightforward.

9 System Settings

A small set of controls lets each library tailor the rules to how it works.

- ◆ **Lending period** — the number of days a book may be kept before it counts as overdue.
- ◆ **Maximum books per member** — how many books one member may hold at the same time.

Changes take effect immediately, and any loans affected by a new setting are re-evaluated automatically.

10 Dashboard & Statistics

The home screen gives an at-a-glance picture of the library's status the moment staff sign in.

- ◆ **Collection size** — how many titles and how many physical copies the library holds.
- ◆ **Currently on loan** — how many books are out right now.
- ◆ **Overdue count** — how many loans need attention.
- ◆ **Membership** — how many members are registered.
- ◆ **Removed books** — how many are recorded as lost, damaged, or withdrawn.
- ◆ **Popularity** — which books are borrowed most often.

11 Search & Filtering

Finding the right record is fast from anywhere in the system.

- ◆ **Search across the catalogue, members, and loans.**
- ◆ **Quick filtering** by the details that matter — book number, name, category, or status.
- ◆ **Sortable lists** so information can be arranged the way staff prefer to read it.

12 Languages

The system speaks the language of its users.

- ◆ **Full interface in English and Romanian.**
- ◆ **Switch languages** without losing your place or your data.
- ◆ **Remembered per person**, so two staff members can each work in their own language.

13 Access, Security & Deployment

The system is built for library staff, and it can be set up to match each library's needs — from a single desk to secure access over the internet.

Staff sign-in. Access is protected by a staff sign-in, so only authorised library employees can view members' details and manage the collection.

Three ways to run the system. The same system can be deployed in whichever way suits a given library, and a library can move from one to another later without losing any data:

On a single computer **SIMPLEST**

The system runs on one computer at the library and is used from there — ideal for a small library operating from a single desk.

On an on-site server, shared across the building **MULTI-DEVICE**

The system runs on a server within the library, and any computer or device on the library's own network can use it — so several staff can work at the same time.

Securely online, accessible from anywhere **ANYWHERE**

The system is hosted online and can be reached from any location through a web browser, protected by staff sign-in — useful when the library needs to be managed off-site.

Data privacy

Each library's information is kept **separate and private** to that library. Member details and borrowing history are never shared between libraries.

14 Business Rules at a Glance

A short summary of the rules the system enforces automatically to keep records trustworthy.

- ◆ Every book copy has a **unique book number** within the active collection.
- ◆ A book copy can be on **only one active loan at a time**.
- ◆ A member may hold **up to a set number of books** at once (a library setting; typically 3).
- ◆ Loans become **overdue automatically** once the lending period passes (a library setting; typically 14 days).
- ◆ Books removed from the collection **always record a reason** and can be **restored**.
- ◆ Books, categories, and members are **archived, never erased**, preserving full history.
- ◆ **Names and book numbers stay unique**, preventing duplicate or conflicting records.

15 Future Enhancements

Features that are natural next steps for the system, available to add as a library's needs grow.

- ◆ **Export to Excel & printable reports** — download catalogue, member, and loan lists for offline use.
- ◆ **Member self-service** — allow members to sign in and view their own current and past loans.
- ◆ **Automated reminders** — email notifications for books that are due soon or overdue.
- ◆ **Reservations** — let members place a hold on a book that is currently on loan.

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